

EDUCATION

University of Guelph

Bachelor of Computing – Computer Science (Co-op), Honours

Sep 2022 – Aug 2027 (Expected)

Guelph, ON

SKILLS

Languages	Python, JavaScript, TypeScript, Java, SQL, C, C++, HTML/CSS
Technologies	Git, Docker, AWS, Node.js, PostgreSQL, Nginx, Unicorn, Redis, ServiceNow, Grafana, Power BI
Frameworks / Libraries	Django, Flask, Express, React, NestJS, SQLAlchemy, Pandas, Selenium, Jest

WORK EXPERIENCE

Software Engineer Intern

Jan 2026 – Present

TD Bank Toronto, ON

- Designed an automated ServiceNow workflow that replaced **weeks-long manual group-linking**, eliminated repetitive configuration work, and standardized eGRC entitlement mapping during organizational changes.
- Built **Python data integration pipelines** connecting ServiceNow, internal systems, and Microsoft SQL Server to support automated eGRC reporting and dashboard inputs.
- Automated ServiceNow script exports to GitHub via Python and the ServiceNow API; added Jest suites covering **80%+** of targeted business logic.
- Developed **JavaScript ServiceNow business rules**, scheduled jobs, and workflow automations for TD's internal risk and compliance operations.

Software Developer Intern

May 2025 – Aug 2025

Health Espresso Inc. Oakville, ON

- Built a **secure real-time messaging microservice** with TypeScript, NestJS, WebSockets, and AWS S3, supporting encrypted practitioner-patient chat, file attachments, read receipts, and role-based access.
- Created a Python Selenium regression suite for critical user/admin flows in the web portal using deterministic test data and explicit waits; cut manual QA by **80%** and caught **15+ pre-release bugs**.
- Built an **AWS Bedrock text-extraction microservice** with API Gateway, Lambda, and PostgreSQL, converting clinical notes into structured patient demographics with **sub-2-second responses**.

Data Engineer Intern

Sep 2024 – Dec 2024

Johnson Electric Hamilton, ON

- Automated ingestion and transformation of factory production data into PostgreSQL with a **Python ETL pipeline**, replacing manual reporting and enabling scheduled processing across manufacturing systems.
- Built **internal Django web applications** to digitize factory-floor workflows, replacing paper-based forms and enabling operators to submit production reports, track issues, and report broken tools through centralized web interfaces.
- Designed **SQL-driven Grafana and Metabase dashboards** to visualize machine utilization, downtime, and production efficiency trends for manufacturing operations monitoring.

PROJECTS

PCTracker | Python, Django, PostgreSQL, Nginx/Gunicorn

pctracker.ca

- Built and deployed a full-stack PC hardware price-tracking platform for **33,000+ components across 8 categories**, with used-market comparison, saved builds, and live build totals on a Linux VPS with Nginx/Gunicorn.
- Integrated the eBay Browse API with OAuth2 for used-market pricing; applied trimmed-mean outlier filtering and listing denylist rules to reject noisy results within eBay's **5,000-call/day** quota.
- Automated weekly catalog refreshes with a **systemd timer**, cycling the full component catalog and persisting historical price snapshots for per-component trend tracking.
- Mapped PassMark CPU/GPU scores onto **6,700+ components** using **fuzzy chip-model matching**, exposing sortable performance data and price-per-benchmark "best value" rankings.

GeoTuring | TypeScript, React, Node.js, Express, Socket.IO, PostgreSQL, Redis, OpenRouter

geoturing.com

- Built and deployed a **real-time social-deduction game** where anonymous users chat 1:1 with either a human or AI agent, then guess whether the opponent was human and identify their geographic origin or AI model family.
- Engineered a **server-authoritative Socket.IO game engine** with matchmaking, automatic AI fallback, turn-based chat rooms, per-turn and round timers, live state synchronization, and session resume across reconnects.
- Designed **OpenRouter-powered AI opponents** across GPT, Claude, Gemini, and Grok with randomized personas, human-like typing cadence, prompt-injection resistance, and counter-guesses that analyze the human player.
- Implemented haversine distance scoring, PostgreSQL persistence for game history and JSONB chat transcripts, **Redis-backed leaderboards**, and privacy protection by rounding IP-derived coordinates to a **~20 km grid**.